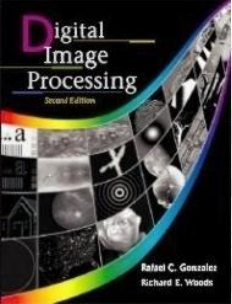


Chapter 11

Representation & Description

- Image segmented into regions, how to represent and describe these regions?
 - 1) In terms of its external characteristics (boundary)
 - 2) In terms of its internal characteristics (pixels in the region)



11.1 Representation – Chain code

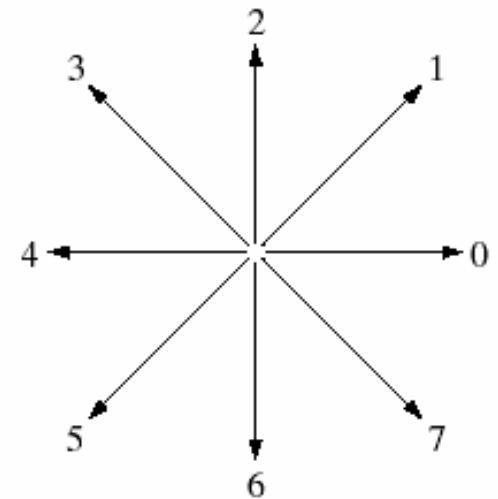
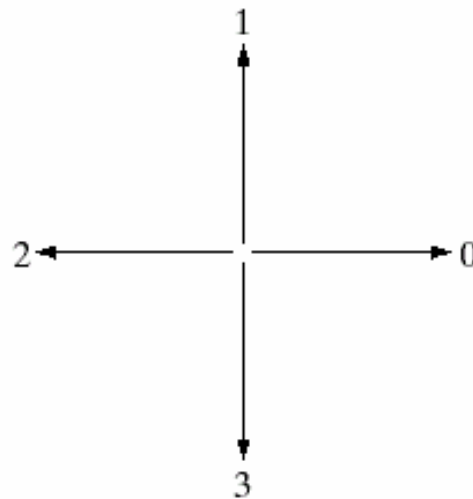
- **Chain codes** are used to represent a **boundary** as a connected sequence of straight line segments of specified length and direction.
- The representation is based on 4- or 8- connectivity.
- Chain code is generated by following a boundary in clockwise direction and assigning a direction to the segments connecting every pair of pixels.
- Disadvantages of chain codes:
 - 1) The chain code is quite long
 - 2) Any small disturbance along the boundary due to noise cause change in the code that may not related to the shape of the boundary.

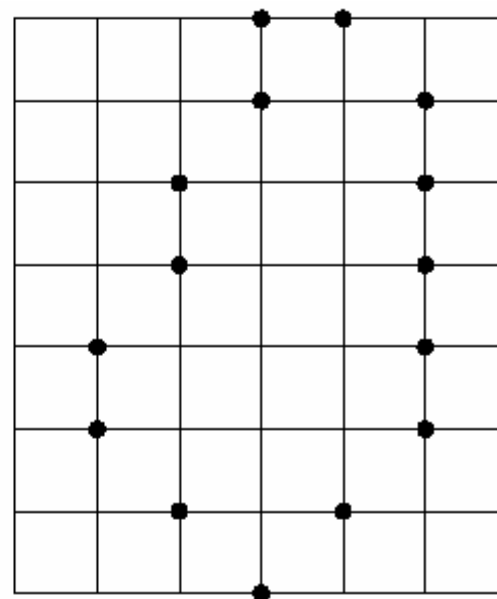
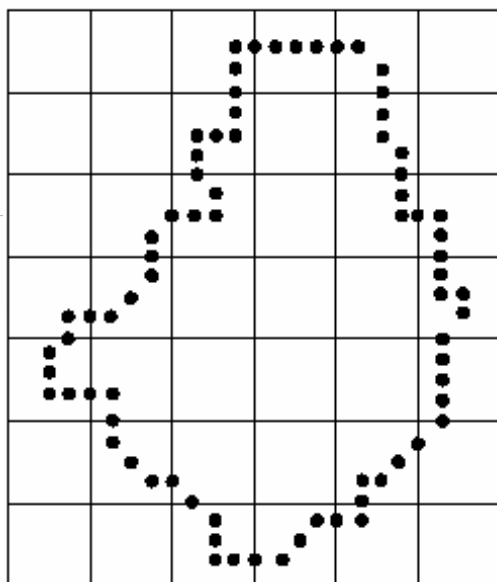
11.1 Representation – Chain code

a b

FIGURE 11.1

Direction numbers for (a) 4-directional chain code, and (b) 8-directional chain code.



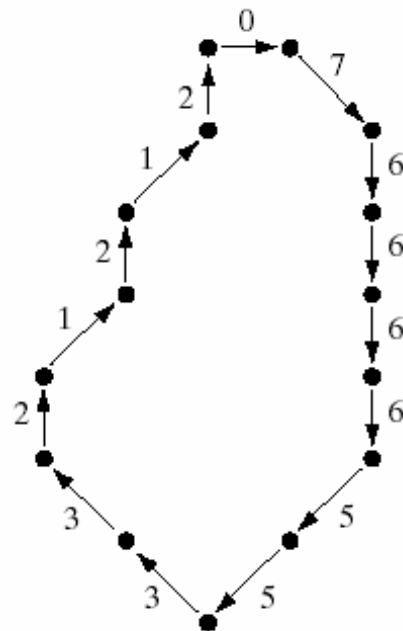
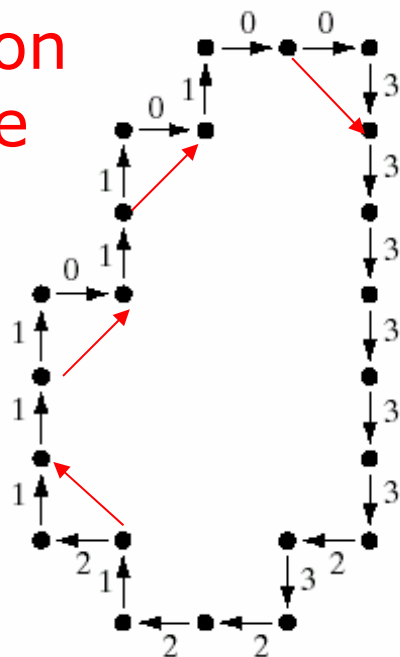


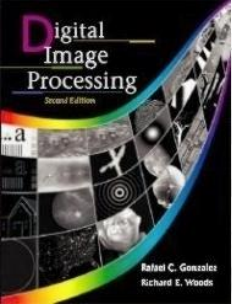
a	b
c	d

FIGURE 11.2
 (a) Digital boundary with resampling grid superimposed.
 (b) Result of resampling.
 (c) 4-directional chain code.
 (d) 8-directional chain code.

11.1

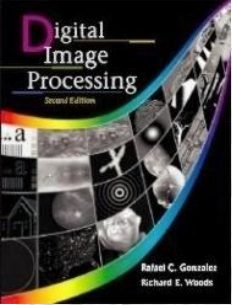
Representation – chain code





11.1 Representation – Chain code

- The chain code of a boundary depends on the ***starting point***.
- ***Normalize*** the chain code by using the first difference of the chain code.
- ***Example***: the chain code is 10103322, the first difference is 3133030 or **3**3133030, the 1st “3” is obtained by connecting the last and the first element of the chain.
- ***Size normalization*** can be obtained by alternating the size of the sampling grid.



11.1 Representation - Polygon approximation

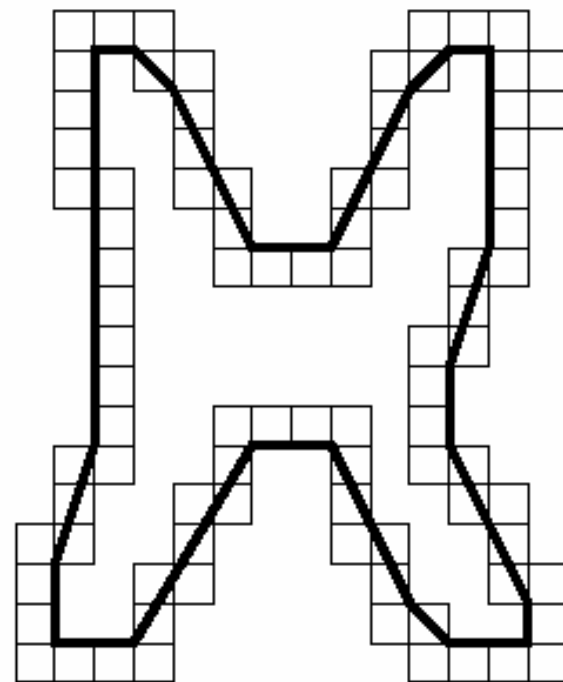
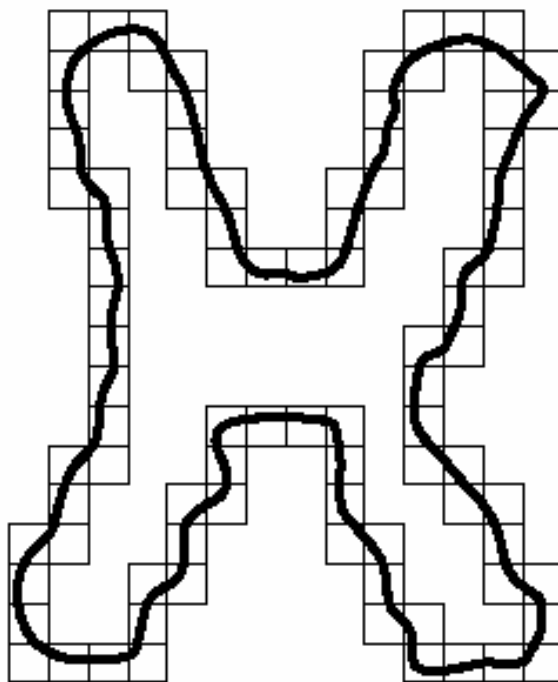
- Minimum perimeter polygons
 - Enclose the boundary by a set of concatenated cells (Fig. 11.3).
 - The enclosure has two walls corresponding to the inside and outside boundaries of the strip of cell.
 - Think of the object boundary as a **rubber band** contained within the wall.
 - The rubber band shrinks and produces a polygon of minimum perimeter that fit the geometry established by the cell strip.

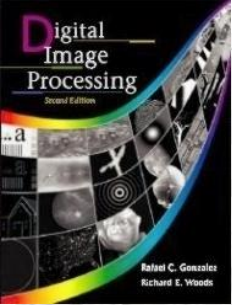
11.1 Representation - Polygon approximation

a b

FIGURE 11.3

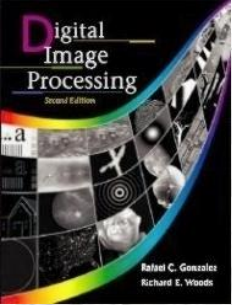
(a) Object boundary enclosed by cells.
(b) Minimum perimeter polygon.





11.1 Representation - Polygon approximation

- Merging technique
 - Merge points along the boundary until the least square error line fit of the points merged so far exceeds a preset threshold.
 - Difficulties: the vertices do not always correspond to inflections (corners) in the original boundary.



11.1 Representation Polygon approximation

- ***Splitting techniques:***
 - Subdivide a segment successively into two parts until a specified criterion is satisfied.
 - The maximum **perpendicular distance** from a boundary segment to the line joining its two end points not exceed a preset threshold.
 - If it does, the farthest point from the line become a vertex, thus subdivide the segment into two sub-segments,
 - This approach has the advantage in seeking prominent inflection points

11.1 Representation - Polygon approximation

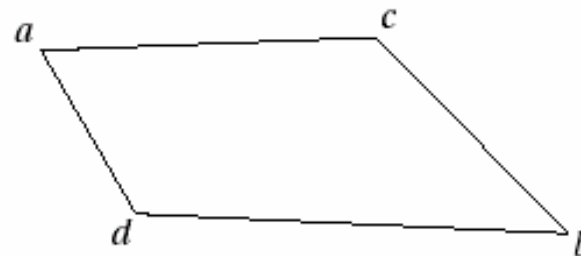
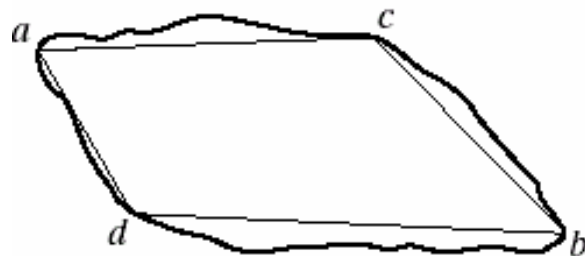
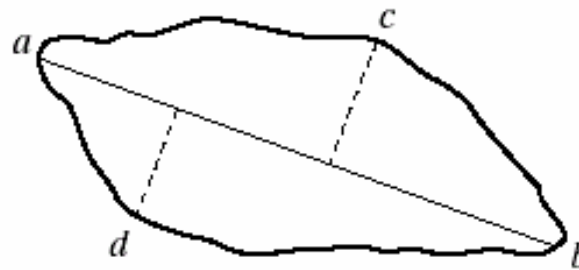
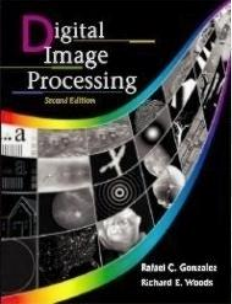


FIGURE 11.4
 (a) Original boundary.
 (b) Boundary divided into segments based on extreme points. (c) Joining of vertices.
 (d) Resulting polygon.



11.1 Representation - Polygon approximation

- **Signature**

- 1-D functional representation of a boundary.

1) Plot the **distance** from the centroid to the boundary as a function of angles (Fig. 11.5), *i. e.*, $r(\theta)$.

- Invariant to translation, but depend on the rotation and scaling.
- Normalizing with respect to rotation.
- Select the starting point as the point farthest to the centroid.

2) Traverse the boundary and plot the **angle** between a line tangent to the boundary at that point and a reference line. Then use the **Slope density function**: (**histogram** of tangent-angle values) as signature.

11.1 Representation - Polygon approximation

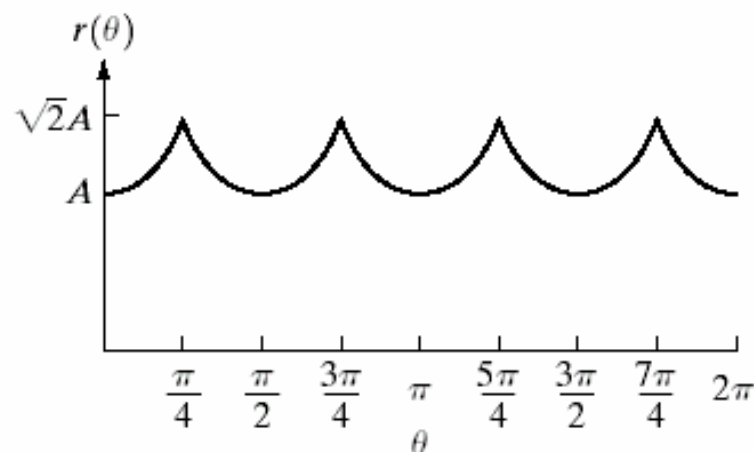
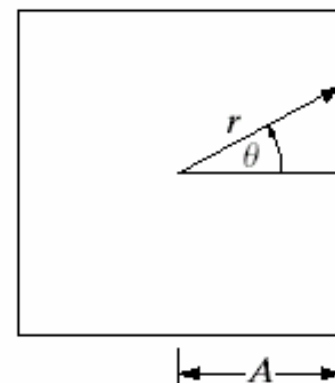
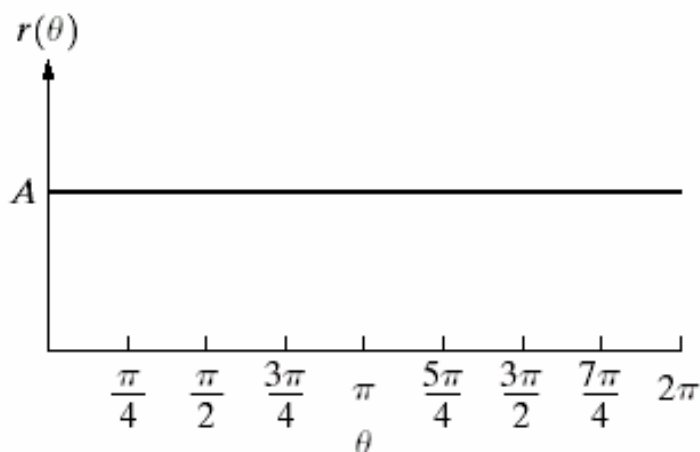
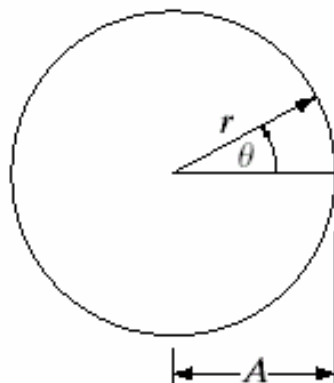
a b

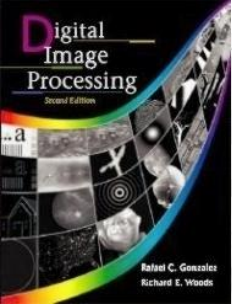
FIGURE 11.5

Distance-versus-angle signatures.

In (a) $r(\theta)$ is constant. In (b), the signature consists of repetitions of the pattern

$r(\theta) = A \sec \theta$ for $0 \leq \theta \leq \pi/4$ and $r(\theta) = A \csc \theta$ for $\pi/4 < \theta \leq \pi/2$.

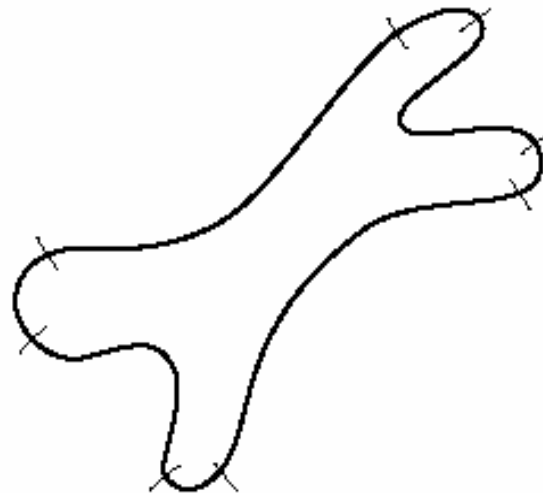
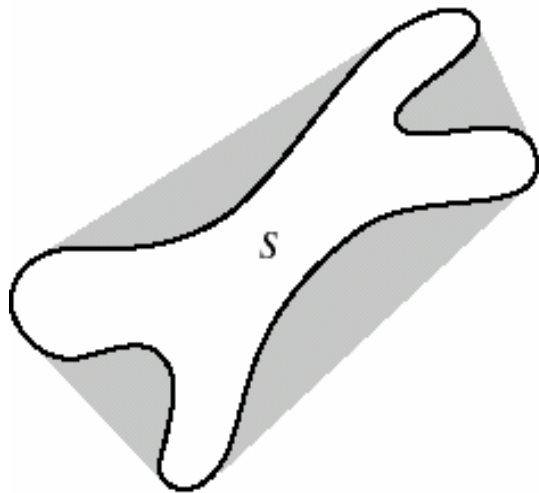




11.1 Representation-boundary segment

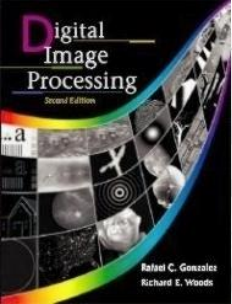
- Convex hull ***H*** of an arbitrary set ***S*** is the smallest convex set containing S .
- The difference $H - S$ is call ***convex deficiency D*** of the set S .
- The region boundary can be partitioned by following the contour of S and marking the points at which a transition is made into or out of a component of the ***convex deficiency***.
- The concept of ***convex hull*** and its ***deficiency*** are equally useful for describing an ***entire region***, as well as just its ***boundary***.

11.1 Representation-boundary segment



a b

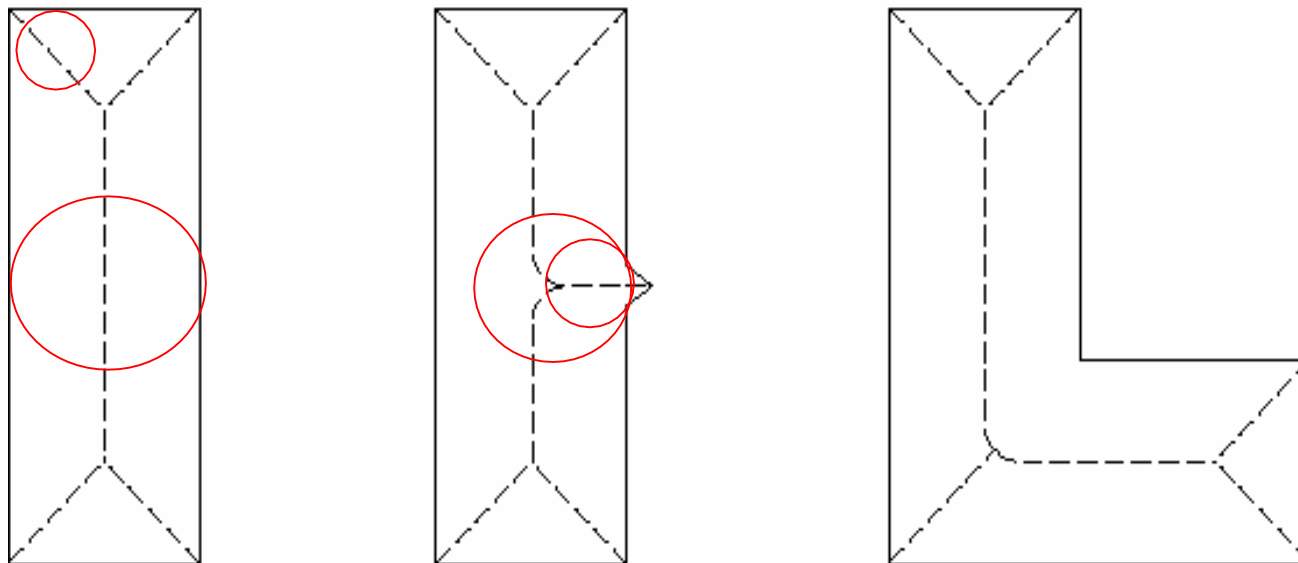
FIGURE 11.6
(a) A region, S , and its convex deficiency (shaded).
(b) Partitioned boundary.



11.1 Representation - Polygon approximation

- ***Skeleton*** of a region can be obtained by ***thinning*** algorithm
- ***Medial axis transformation (MAT)***:
 - 1) For each point in region ***R***, we find its **closest neighbor** in border ***B***.
 - 2) If ***p*** has more than one such neighbor, it is said to belong to the medial axis (skeleton) of ***R***.
- ***Thinning algorithm***: iteratively delete the edge points of a region subject to
 - 1) Does not remove the end points
 - 2) Does not break connectivity
 - 3) Does not cause excessive erosion of the region.

11.1 Representation - Polygon approximation



a b c

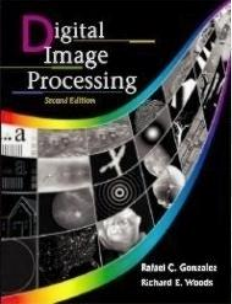
FIGURE 11.7
Medial axes
(dashed) of three
simple regions.

11.1 Representation - Polygon approximation

p_9	p_2	p_3
p_8	p_1	p_4
p_7	p_6	p_5

FIGURE 11.8

Neighborhood arrangement used by the thinning algorithm.



11.1 Representation - Polygon approximation

- **Thinning algorithm**

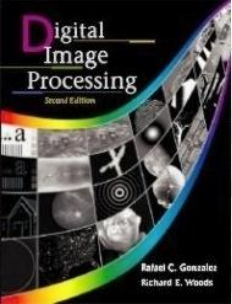
Step 1) flag a contour point p_1 for deletion if the following conditions are satisfied:

- a) $2 \leq N(p_1) \leq 6$, where $N(p_1)$ is the number of neighbors of p_1 .
- b) $T(p_1) = 1$, where $T(p_1)$ is number of 0-1 transitions in the ordered sequence $p_2, p_3, \dots, p_8, p_9, p_2$
- c) $p_2 \bullet p_4 \bullet p_6 = 0$
- d) $p_4 \bullet p_6 \bullet p_8 = 0$

If all conditions are satisfied, the point is flagged for deletion.

Step 2) Conditions (c) and (d) changed to

- c') $p_2 \bullet p_4 \bullet p_8 = 0$
- d') $p_2 \bullet p_6 \bullet p_8 = 0$



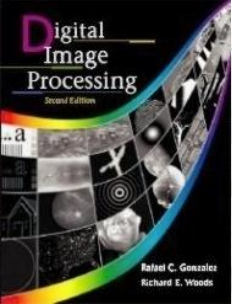
11.1 Representation - Polygon approximation

- **Thinning algorithm**

- 1) Apply step 1 to flag border points for deletion
- 2) Deleting the flagged point
- 3) Apply step 2 to flag the remaining border points for deletion.
- 4) Delete the flagged points

The basic procedure is applied iteratively until no further points are deleted.

- Condition (a) is violated when p_1 is the end point of a skeleton stroke.
- Condition (b) is violated when it is applied to points on stroke 1 pixel thick.



11.1 Representation - Polygon approximation

FIGURE 11.9

Illustration of conditions (a) and (b) in Eq. (11.1-1). In this case $N(p_1) = 4$ and $T(p_1) = 3$.

0	0	1
1	p_1	0
1	0	1

11.1 Representation - Polygon approximation

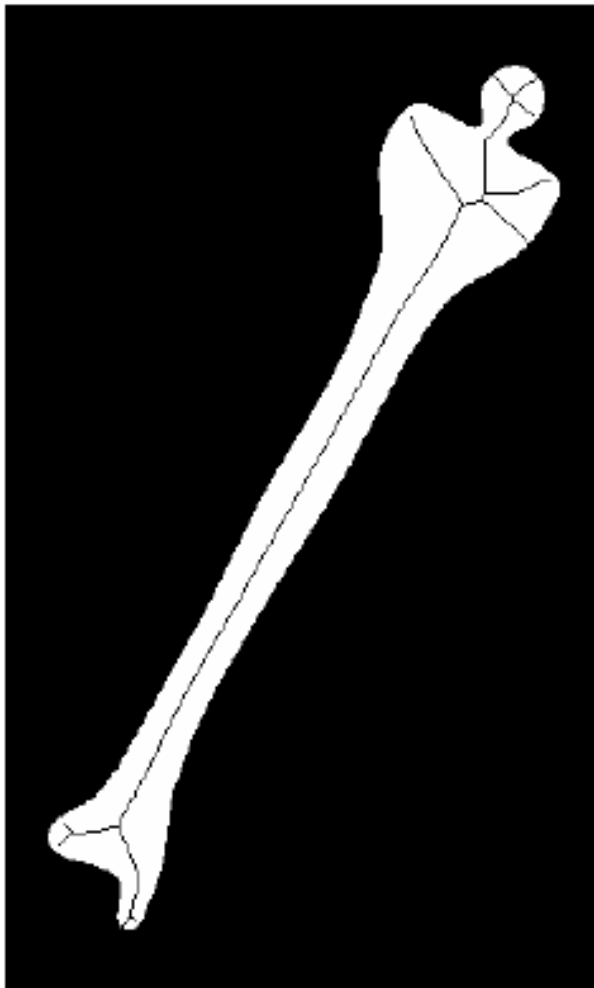
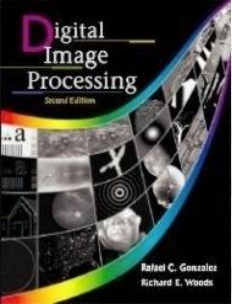


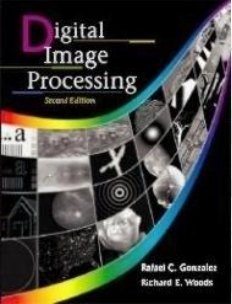
FIGURE 11.10
Human leg bone
and skeleton of
the region shown
superimposed.



11.2 Boundary descriptor

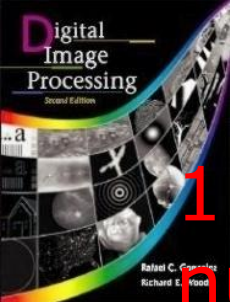
- Simple descriptors

- 1) Length
- 2) Diameter: $\text{Diam}(B) = \max[D(p_i, p_j)]$ where p_i and p_j are points on the boundary.
- 3) Major axis and minor axis
- 4) Basic rectangle
- 5) Eccentricity = major axis/minor axis
- 6) Curvature: changes of slope.
- 7) Point p belongs to a segment which is convex if the change of slope at p is nonnegative and concave otherwise.
- 8) P is a *corner* depends on the curvature.



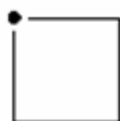
11.2 Boundary Description - shape number

- The ***first difference*** of a chain-coded boundary depends on the starting point.
- The ***shape number*** of a chain coded boundary is defined as the first difference of smallest magnitude.
- The ***difference*** of a chain code is independent of its rotation, it depends on the orientation of the grid.
- The order ***n*** of a shape number is defined as the number of digits in its representation.



11.2 Boundary Description- shape number

Order 4

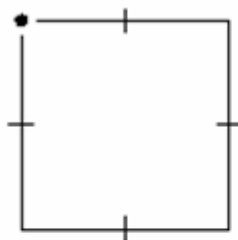


Chain code: 0 3 2 1

Difference: 3 3 3 3

Shape no.: 3 3 3 3

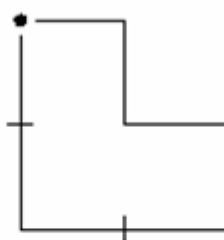
Order 8



Chain code: 0 0 3 3 2 2 1 1

Difference: 3 0 3 0 3 0 3 0

Shape no.: 0 3 0 3 0 3 0 3



Chain code: 0 3 0 3 2 2 1 1

Difference: 3 3 1 3 3 0 3 0

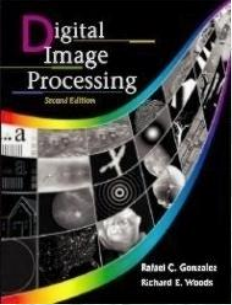
Shape no.: 0 3 0 3 3 1 3 3



Chain code: 0 0 0 3 2 2 2 1

Difference: 3 0 0 3 3 0 0 3

Shape no.: 0 0 3 3 0 0 3 3

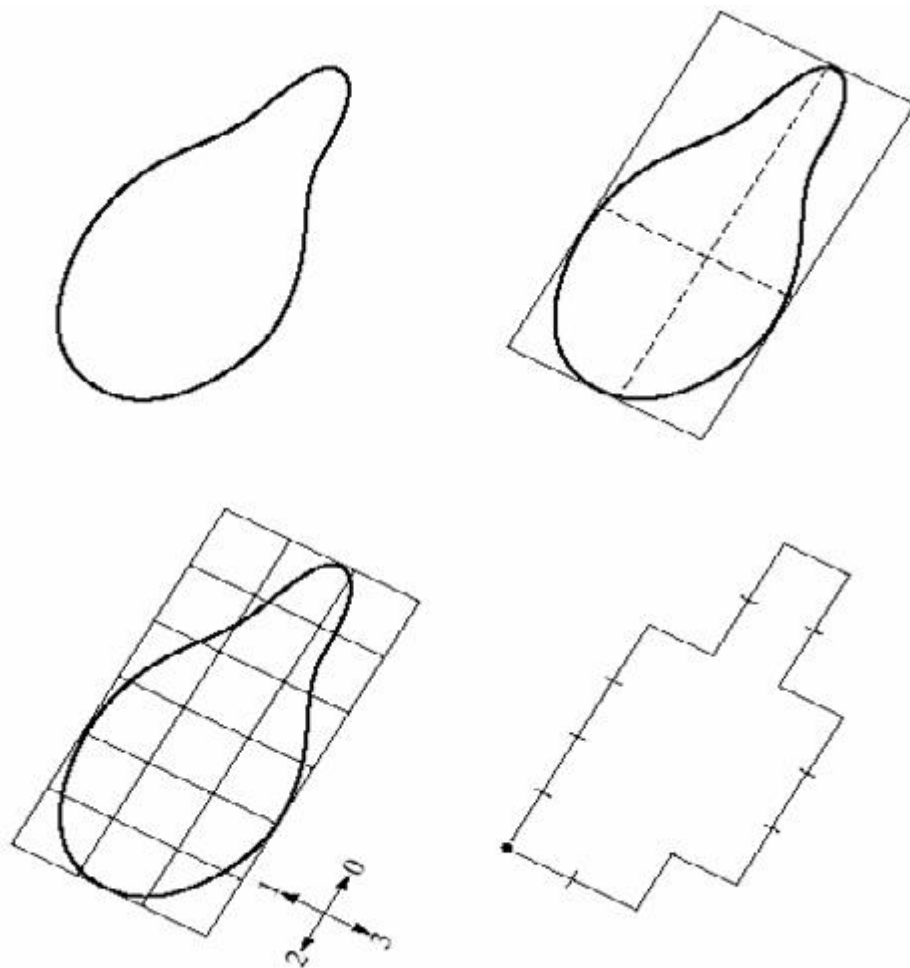


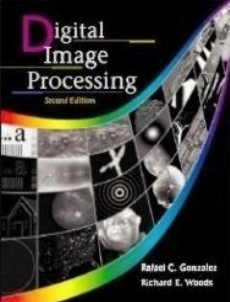
11.2 Boundary Description- shape number

- Example (Fig. 11.12)
- 1. Find the basic rectangle for $n=18$ (boundary)
- 2. Find the major and minor axis
- 3. Find the closest rectangle of order 18 is 3×6
- 4. obtain chain code
- 5. find the difference
- 6. find the shape no.

a b
 c d

FIGURE 11.12
 Steps in the
 generation of a
 shape number.

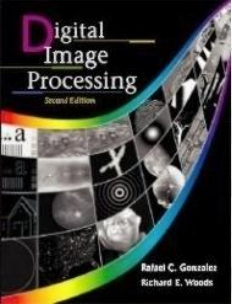




Chain code: 0 0 0 0 3 0 0 3 2 2 3 2 2 2 1 2 1 1

Difference: 3 0 0 0 3 1 0 3 3 0 1 3 0 0 3 1 3 0

Shape no.: 0 0 0 3 1 0 3 3 0 1 3 0 0 3 1 3 0 3



11.2 Boundary Description–Fourier Descriptor

- For a K -point digital boundary, starting at an arbitrary point (x_0, y_0) , K coordinate pairs (x_0, y_0) , (x_{01}, y_{01}) , $\dots, (x_{K-1}, y_{K-1})$ are encountered in counterclockwise direction.
- Let $s(k)=[x(k), y(k)]$ for $k=0, 1, \dots, K-1$,
or $s(k)=x(k)+jy(k)$
- The 1-D DFT of $s(k)$ is
$$a(u) = \frac{1}{K} \sum_{k=0}^{K-1} s(k) e^{-j2\pi uk/K}$$
- The inverse DFT of $a(u)$ is
$$s(k) = \frac{1}{K} \sum_{u=0}^{K-1} a(u) e^{j2\pi uk/K}$$

11.2 Boundary Description- Fourier Descriptor

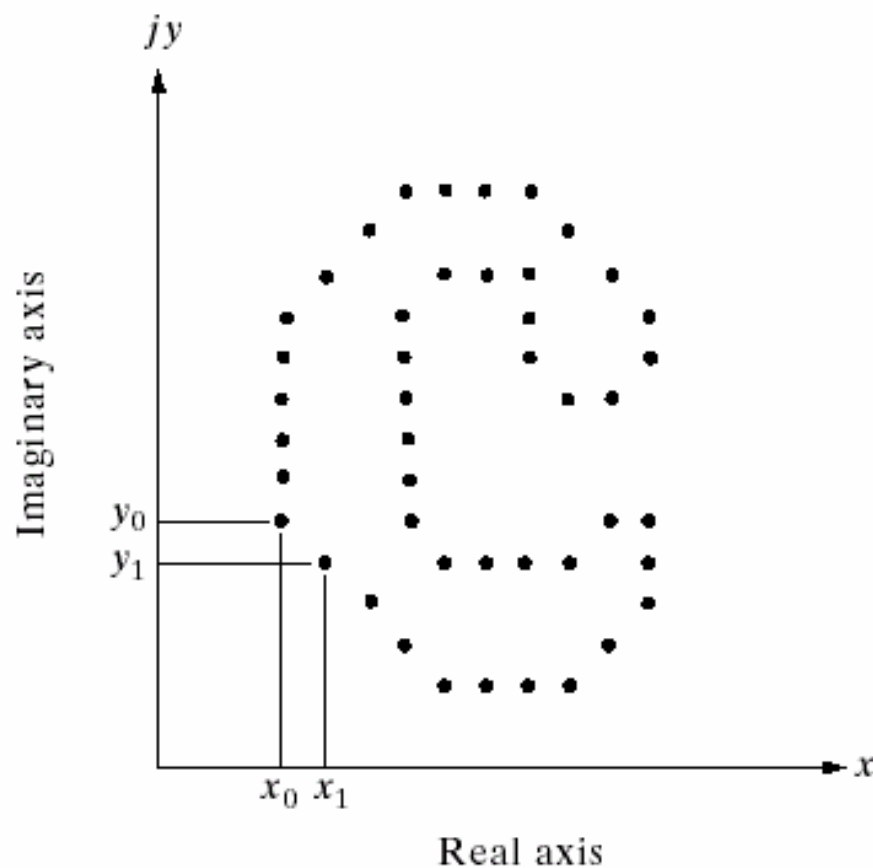
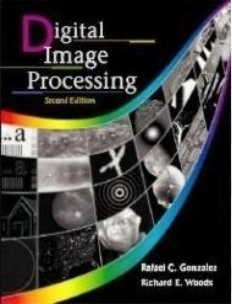


FIGURE 11.13 A digital boundary and its representation as a complex sequence. The points (x_0, y_0) and (x_1, y_1) shown are (arbitrarily) the first two points in the sequence.



11.2 Boundary Description - Fourier Descriptor

- If only the first P coefficients ($P < K$) are used then

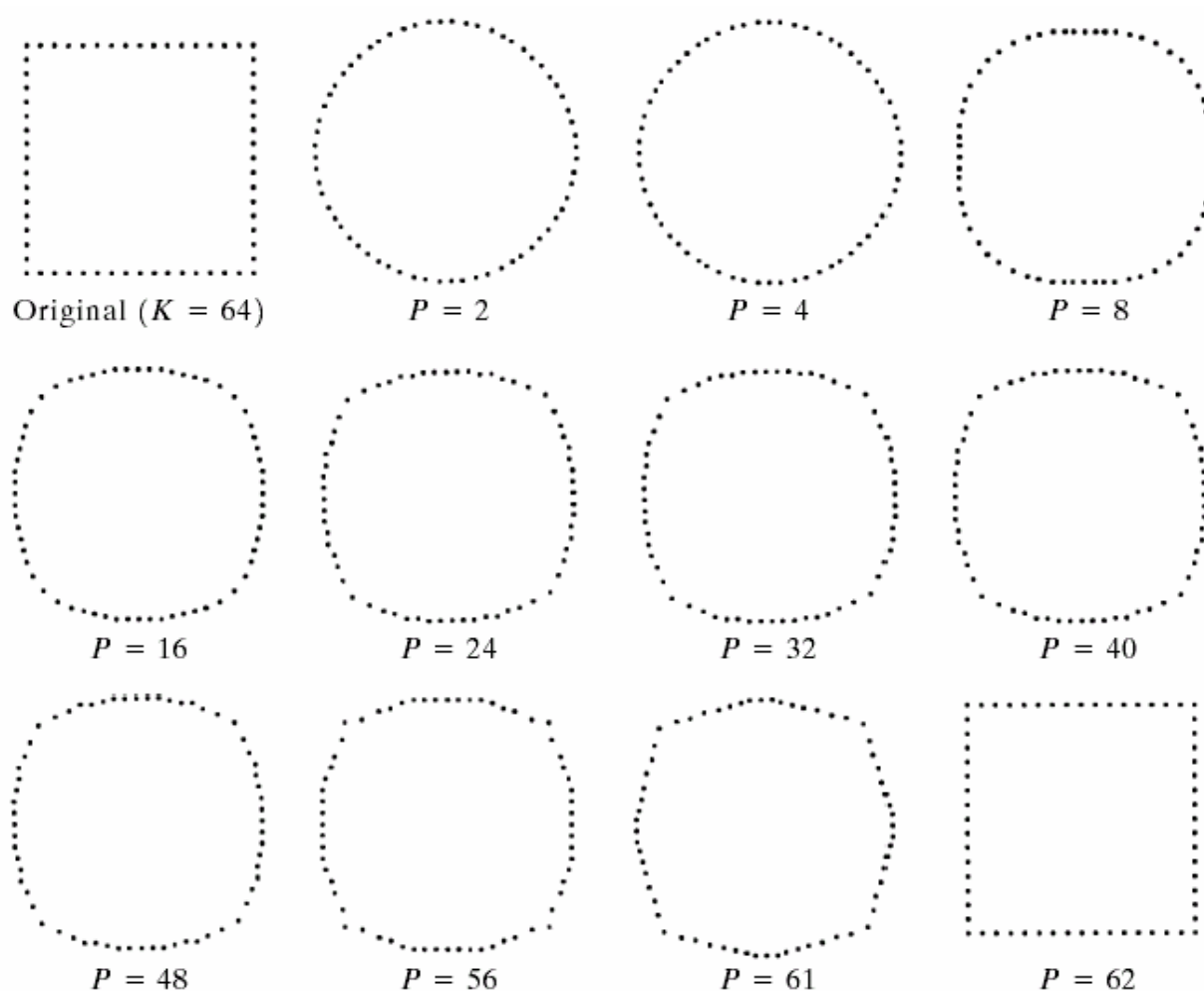
$$\hat{s}(k) = \sum_{u=0}^{P-1} a(u) e^{j2\pi uk/K}$$

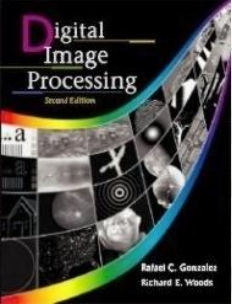
- The coefficients $\{a(u)\}$ carry shape information which are insensitive to translation, rotation, and scale change of the shape.
- The descriptors are insensitive to the change of starting point.
- Rotation of a point by an angle θ about the origin of the complex plane is accomplished by multiplying the point by $e^{j\theta}$.
- The rotated sequence $s(k)e^{j\theta}$ whose Fourier descriptors are

$$a_r(u) = \frac{1}{K} \sum_{k=0}^{K-1} s(k) e^{j\theta} e^{-j2\pi uk/K} = a(u) e^{j\theta}$$

11.2 Boundary Description - Fourier-Descriptor

FIGURE 11.14
Examples of reconstruction from Fourier descriptors. P is the number of Fourier coefficients used in the reconstruction of the boundary.





11.2 Boundary Description -Fourier-Descriptor

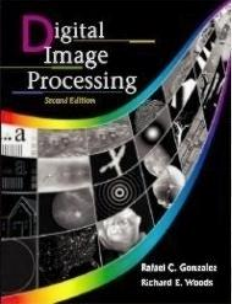
Transformation	Boundary	Fourier Descriptor
Identity	$s(k)$	$a(u)$
Rotation	$s_r(k) = s(k)e^{j\theta}$	$a_r(u) = a(u)e^{j\theta}$
Translation	$s_t(k) = s(k) + \Delta_{xy}$	$a_t(u) = a(u) + \Delta_{xy}\delta(u)$
Scaling	$s_s(k) = \alpha s(k)$	$a_s(u) = \alpha a(u)$
Starting point	$s_p(k) = s(k - k_0)$	$a_p(u) = a(u)e^{-j2\pi k_0 u/K}$

TABLE 11.1
Some basic
properties of
Fourier
descriptors.

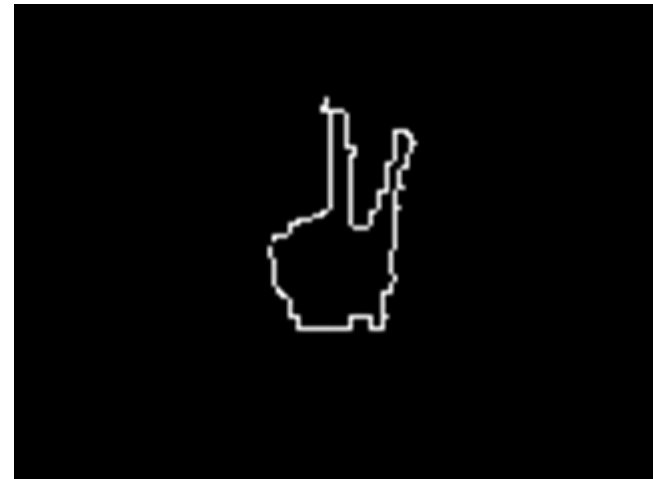
1) Translation: $s_t(k) = s(k) + \Delta_{xy} = [x(k) + \Delta x] + j[y(k) + \Delta y]$

2) Change the starting point of the sequence to $k=k_0$ from $k=0$ as

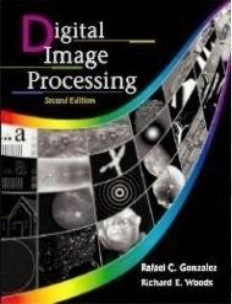
$$s_p(k) = s(k - k_0) = x(k - k_0) + j y(k - k_0)$$



11.2 Boundary Description – Fourier Descriptor



The contour of hand silhouette.



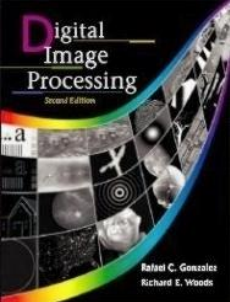
11.2 Fourier Descriptor

- Fourier series of a sequence of points $\{x(m), y(m)\}$ can be defined as

$$x(m) = \sum a(n) e^{j2\pi nm / N} \quad y(m) = \sum b(n) e^{j2\pi nm / N}$$

where $a(n)$ and $b(n)$ are the Fourier coefficient

$$a(n) = \sum_{m=1}^N x(m) e^{-j2\pi nm / N} \quad b(n) = \sum_{m=1}^N y(m) e^{-j2\pi nm / N}$$



11.2 Fourier Descriptor

- Assuming local variation of hand shape is smooth so that the higher order terms of the Fourier descriptor are not necessary.
- To normalize the size of hand gesture we let $S(n)=r(n)/r(1)$ (normalization), and we have

$$r(n) = \sqrt{|a(n)|^2 + |b(n)|^2} \quad n=1,2,\dots,22$$

- Using 22 harmonics of the FD's coefficient, $S(n)$, is enough to describe the macroscopic information of the hand shape.
- FD is translation, rotation, and scaling invariance.

11.2 Fourier Descriptor



(a) Original image



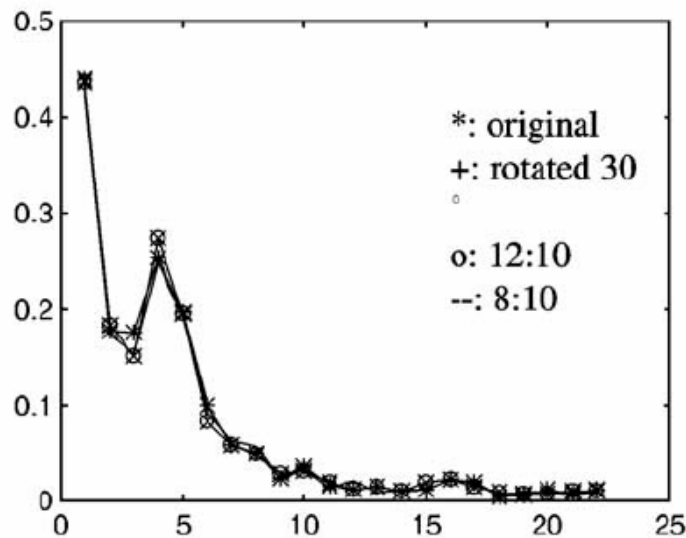
(b) Rotation 90°



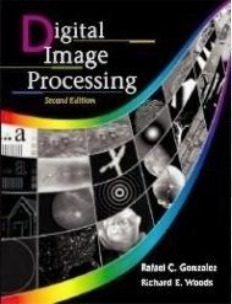
(c) Zoom in 12:10



(d) Zoom out 8:10



(e) Fourier descriptor vectors of the four different shapes without the first term.



11.2 Boundary Description -Statistical moment

- The shape of boundary segments can be described quantitatively by using simple statistical moments such as *mean*, *variance*, and *higher-order moments*.
- Figure 11.5 represented as 1-D function $g(r)$.
- Treat the amplitude of g as a discrete random variable v and form an amplitude histogram $p(v_i)$, $i=0, 1, \dots, A-1$, where A is the number of discrete amplitude increments in which we divide the amplitude scale.

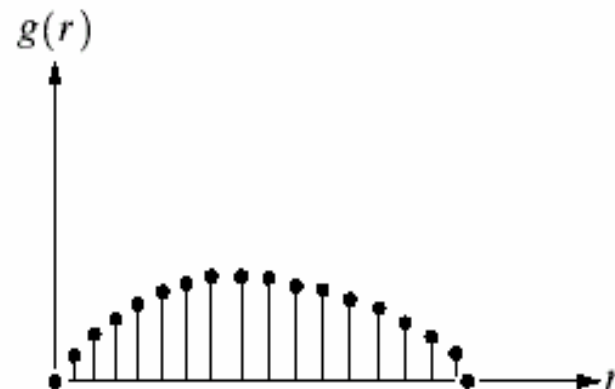
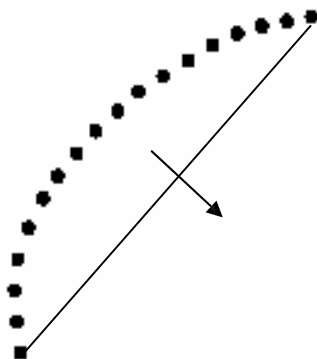
11.2 Boundary Description -Statistical moment

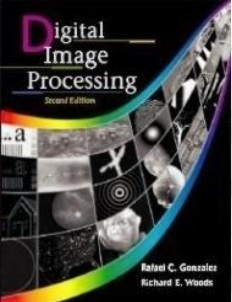
a b

FIGURE 11.15

(a) Boundary segment.

(b) Representation as a 1-D function.





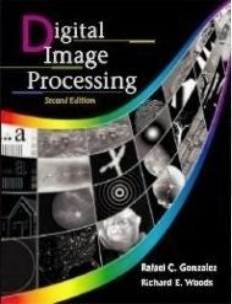
11.2 Boundary Description -Statistical moment

- The n th moment of v about its mean m is

$$\mu_n(v) = \sum_{i=0}^{A-1} (v_i - m)^n p(v_i)$$

where the mean is $m = \sum_{i=0}^{A-1} v_i p(v_i)$

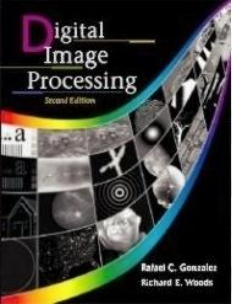
- The m is the mean and μ_2 is the variance.



11.2 Boundary Description -Statistical moment

- An alternative approach is normalize $g(r)$ to unit area and treat it as histogram.
- $g(r_i)$ is treated as the probability of value r_i occurring.
- The moments are
$$\mu_n(r) = \sum_{i=0}^{K-1} (r_i - m)^n g(r_i)$$

where
$$m = \sum_{i=0}^{K-1} r_i g(r_i)$$



11.3 Regional Descriptors -Simple Descriptor

- Area is the number of pixels in the regions
- Perimeter is the length of the boundary.
- Compactness= $(\text{perimeter})^2/\text{area}$.

11.3 Regional Descriptors

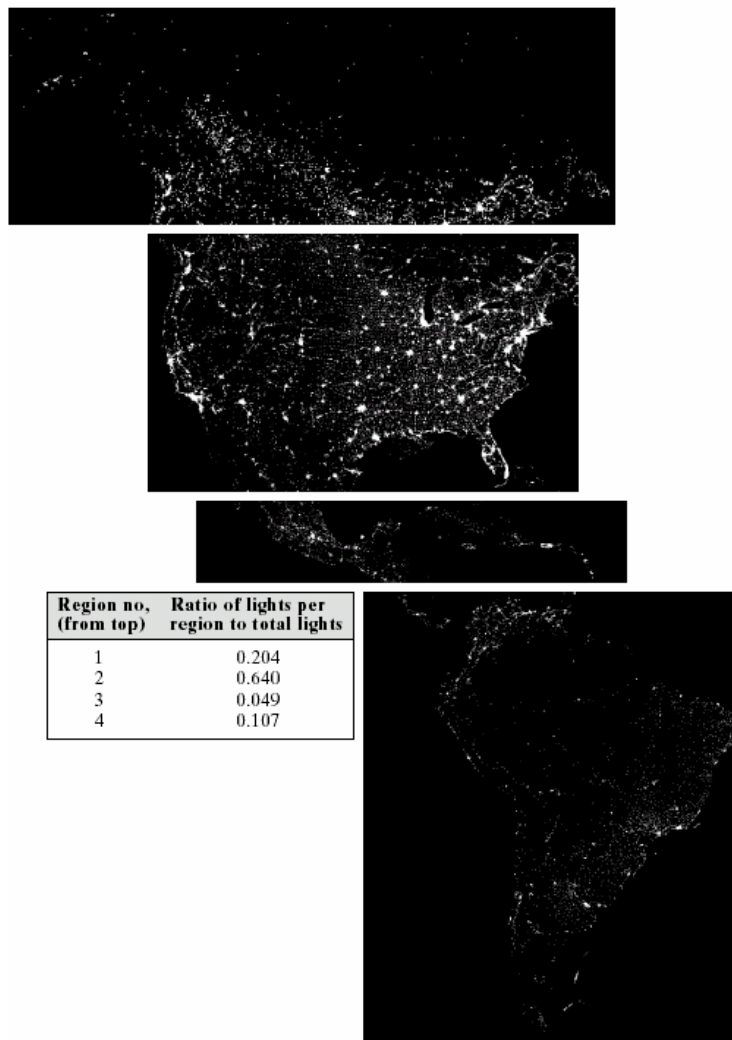
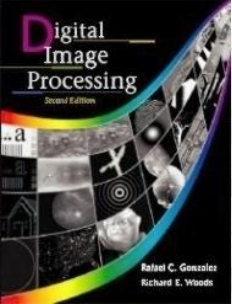


FIGURE 11.16 Infrared images of the Americas at night. (Courtesy of NOAA.)



11.3 Regional Descriptors

-Topological Descriptor

- Topology is the study of properties of a figure that are unaffected by any deformation (rubber-sheet distortion).
- The number of *holes*: H
- The number of *connected components*: C
- *Euler number* E : $E = C - H$.

11.3 Regional Descriptors -Topological Descriptor

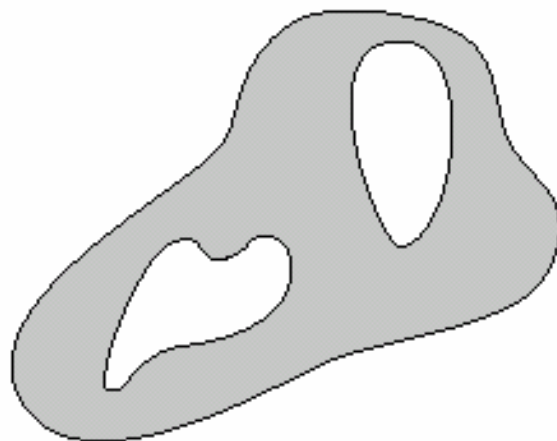


FIGURE 11.17 A region with two holes.

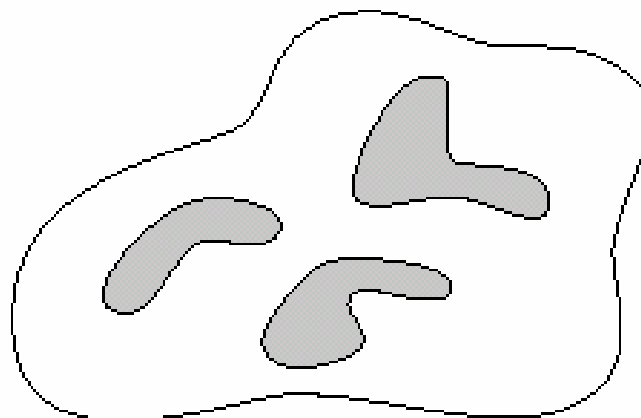
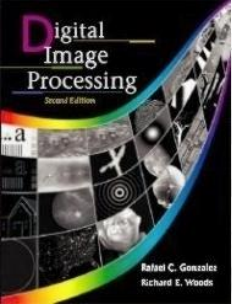


FIGURE 11.18 A region with three connected components.



11.3 Regional Descriptors -Topological Descriptor

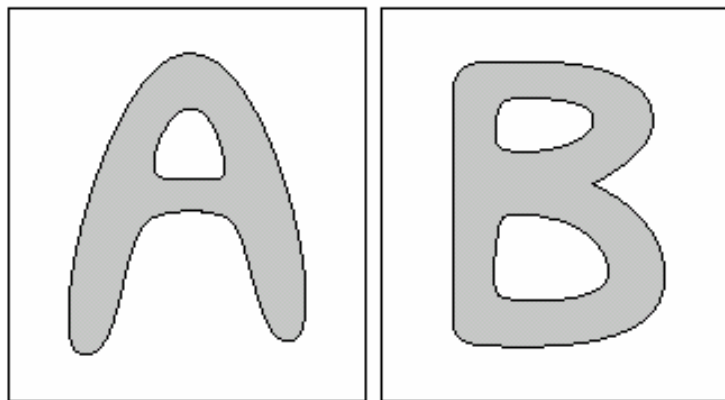
- Regions represented by straight-line segments (polygonal networks), such as Fig. 11.20, has the following relationship in topology as

$$E = V - Q + F = C - H$$

where V is the number of vertices and Q is the number of edges.

- Segmentation is based on the thresholding.
- How the connected components can be used to “finish” the segmentation.
- Figure 11.21(b) has 1591 connected components, $C=1591$, and its Euler number $E=1552$, and $H=39$. Figure 11.21(c) shows the connected component with 8479 elements

11.3 Regional Descriptors -Topological Descriptor



a b

FIGURE 11.19 Regions with Euler number equal to 0 and -1 , respectively.

$$V - Q + F = C - H = E$$

$$7 - 11 + 2 = 1 - 3 = -2$$

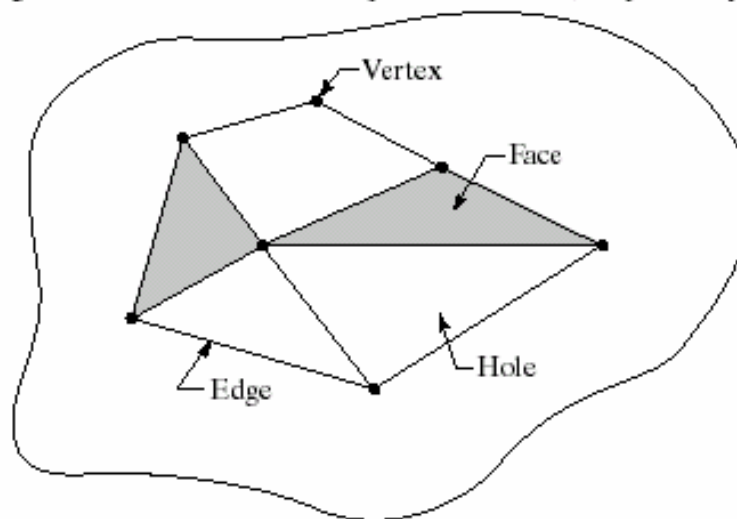
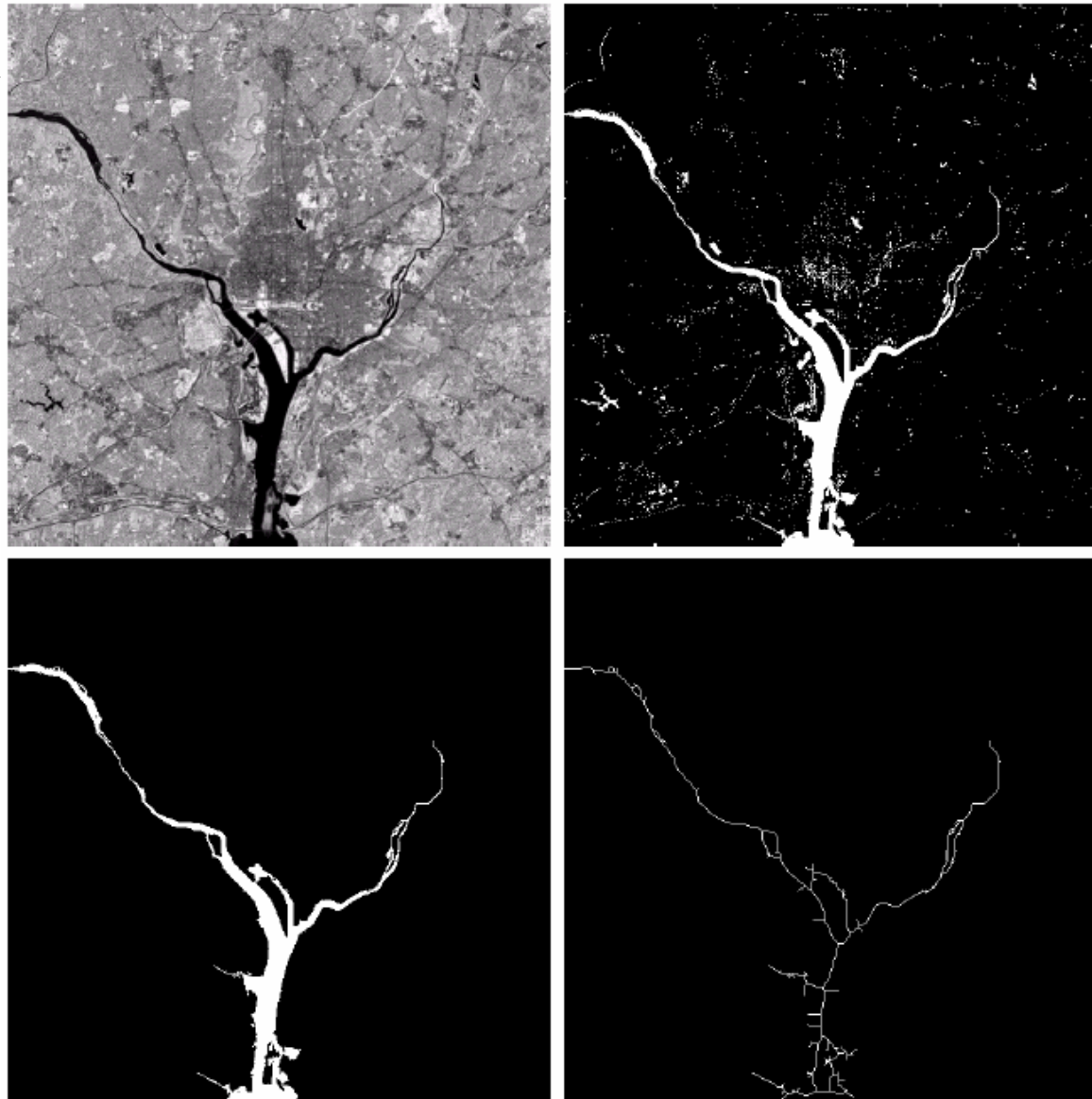


FIGURE 11.20 A region containing a polygonal network.

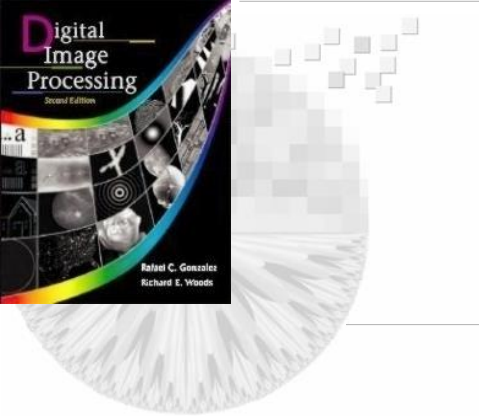
11.3 Regional Descriptors -Topological Descriptor



a	b
c	d

FIGURE 11.21

(a) Infrared image of the Washington, D.C. area. (b) Thresholded image. (c) The largest connected component of (b). (d) Skeleton of (c).



Thanks